

Chern–Simons Phase Selection on $L(6, 1)$: Differential-Cohomological Result and Equivariant-Lift Obstruction

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Abstract

We isolate the level-selection step of Information-Topological Theory (ITT) and formulate it in a mathematically conservative way. Starting from the lens-space sector $L(6, 1) = S^3/\mathbb{Z}_6$ and the convention-fixed flat representation ρ_1 , we use the differential Chern–Simons refinement of the integral level- k class kc_2 to derive the closed-manifold phase

$$Z_k(L(6, 1), \rho_1) = \exp(2\pi i k \text{cs}(\rho_1)) = e^{-2\pi i k/6}.$$

This phase canonically induces, relative to the selected generator of $\mathbb{Z}_6$, a cyclic $U(1)$ character. We then analyze whether the same phase can be identified with the action of that generator on a hypothetical one-dimensional equivariant fiber over the three-dimensional connection sector. Two independent obstructions are found. First, the standard Chern–Simons line-bundle construction is a transgression to even-dimensional manifolds; the three-dimensional closed-manifold theory supplies a phase rather than an ordinary prequantum line over the three-dimensional connection space. Second, whenever a one-dimensional equivariant linearization exists, its fiber character is ambiguous up to $\text{Hom}(\mathbb{Z}_6, U(1)) \cong \mu_6$ unless an additional normalization datum is specified. Thus the canonical identification of the fiber character with the lens-space Chern–Simons phase does not follow from the presently specified data.

We therefore replace the unresolved equivariant-line identification by the explicit ITT phase-triviality selection postulate $Z_k(L(6, 1), \rho_1) = 1$. Under this postulate and ordinary integer level quantization, the minimal positive level is derived as $k_{\min} = 6$. The result is not presented as a universal theorem of ordinary $SU(2)$ Chern–Simons theory.

1 Introduction and Logical Status

The purpose of this paper is to isolate one narrow mathematical issue in Information-Topological Theory (ITT): how the lens-space sector $L(6, 1) = S^3/\mathbb{Z}_6$ constrains the Chern–Simons level.

The foundational formulation distinguishes ordinary Chern–Simons quantization,

$$k \in \mathbb{Z}, \tag{1}$$

from the ITT-specific selection of a minimal level. The latter is based on the fundamental flat sector

$$\rho_1 : \pi_1(L(6, 1)) \cong \mathbb{Z}_6 \longrightarrow SU(2), \tag{2}$$

with

$$\rho_1(\gamma) = \begin{pmatrix} e^{2\pi i/6} & 0 \\ 0 & e^{-2\pi i/6} \end{pmatrix} \tag{3}$$

and convention-fixed Chern–Simons invariant

$$\text{cs}(\rho_1) = -\frac{1}{6} \pmod{\mathbb{Z}}. \quad (4)$$

Earlier formulations attempted to identify

$$\exp(2\pi i k \text{cs}(\rho_1)) \quad (5)$$

directly with a stabilizer character on a prequantum line fiber over a three-dimensional connection space. The present paper shows that this identification is not a theorem of the currently specified data. The correct derived result is the Chern–Simons phase itself, together with a precisely stated obstruction to promoting it canonically to the desired fiber action.

2 Geometric Input

2.1 The lens space and its deck group

Let

$$\Gamma = \mathbb{Z}_6 = \langle \zeta \mid \zeta^6 = 1 \rangle, \quad \omega = e^{2\pi i/6}. \quad (6)$$

The free action on

$$S^3 = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 = 1\} \quad (7)$$

is

$$\delta_\zeta(z_1, z_2) = (\omega z_1, \omega z_2). \quad (8)$$

Hence

$$M := S^3/\Gamma = L(6, 1), \quad \pi_1(M) \cong \Gamma. \quad (9)$$

2.2 The selected flat representation

Fix

$$\rho_1(\zeta) = \begin{pmatrix} \omega & 0 \\ 0 & \omega^{-1} \end{pmatrix} \in SU(2). \quad (10)$$

The associated principal bundle can be realized equivariantly on the universal cover as

$$\tilde{P} = S^3 \times SU(2) \quad (11)$$

with left Γ -action

$$\Phi_\gamma(x, h) = (\delta_\gamma x, \rho_1(\gamma)^{-1} h) \quad (12)$$

and the usual right $SU(2)$ action on the second factor. The quotient

$$\boxed{P_{\rho_1} = \tilde{P}/\Gamma} \quad (13)$$

is a principal $SU(2)$ bundle over $L(6, 1)$.

The trivial connection on $\tilde{P}$ is Γ -invariant and descends to a flat connection A_{ρ_1} with holonomy ρ_1 . The non-trivial information is therefore carried by the quotient/equivariant descent data, not by a non-trivial flat connection on simply connected S^3 .

3 The Γ -Equivariant Connection Groupoid

Let

$$\mathcal{A}(S^3) = \Omega^1(S^3, \mathfrak{su}(2)). \quad (14)$$

The connections on $\tilde{P}$ that descend to P_{ρ_1} are exactly

$$\boxed{\mathcal{A}_{\rho_1}^\Gamma = \left\{ A \in \Omega^1(S^3, \mathfrak{su}(2)) : \delta_\gamma^* A = \text{Ad}_{\rho_1(\gamma)^{-1}} A, \quad \gamma \in \Gamma \right\}.} \quad (15)$$

The corresponding equivariant gauge group is

$$\boxed{\mathcal{G}_{\rho_1}^\Gamma = \left\{ u : S^3 \rightarrow SU(2) : u(\delta_\gamma x) = \rho_1(\gamma)^{-1} u(x) \rho_1(\gamma), \quad \gamma \in \Gamma \right\}.} \quad (16)$$

With

$$A^u = u^{-1} A u + u^{-1} du, \quad (17)$$

we define the connection groupoid

$$\boxed{\mathfrak{C}_{\rho_1} := \left[\mathcal{A}_{\rho_1}^\Gamma / \mathcal{G}_{\rho_1}^\Gamma \right].} \quad (18)$$

The distinguished flat object is the class of the trivial connection on the covering space,

$$[A_0], \quad A_0 = 0, \quad (19)$$

together with its non-trivial equivariant lift determined by ρ_1 .

Remark 3.1. *The groupoid above is not restricted to the discrete flat moduli space. Continuous auxiliary paths relevant to spectral-flow or transgression questions live in the full equivariant connection space. This avoids identifying a continuous interpolation with a path inside a discrete set of flat equivalence classes.*

4 Differential Chern–Simons Data

4.1 Integral characteristic class

At integer level k , the relevant characteristic class is

$$kc_2 \in H^4(BSU(2); \mathbb{Z}). \quad (20)$$

Its differential refinement is a degree-four Cheeger–Simons differential character, denoted schematically by

$$\hat{c}_{2,k} \in \hat{H}^4(BSU(2); \mathbb{Z}). \quad (21)$$

Differential characters refine integral characteristic classes by retaining both curvature and secondary holonomy data[1].

Pulling the differential class back along the classifying data of (P_{ρ_1}, A_{ρ_1}) gives

$$\hat{C}_k(P_{\rho_1}, A_{\rho_1}) \in \hat{H}^4(L(6, 1); \mathbb{Z}). \quad (22)$$

4.2 Three-dimensional evaluation

For a closed oriented three-manifold, a degree-four differential character evaluates on the fundamental three-cycle and produces a $U(1)$ phase. In the Chern–Simons case this evaluation is the exponentiated Chern–Simons invariant[2, 1]:

$$\boxed{Z_k(M, A) = \text{Hol}_M \left(\widehat{C}_k(P, A) \right) = \exp \left(2\pi i k \text{cs}(A) \right).} \quad (23)$$

For the selected lens-space sector,

$$\text{cs}(\rho_1) = -\frac{1}{6} \pmod{\mathbb{Z}}, \quad (24)$$

so

$$\boxed{Z_k(L(6, 1), \rho_1) = e^{-2\pi i k/6}.} \quad (25)$$

Proposition 4.1 (Differential-CS phase). *The phase*

$$Z_k := Z_k(L(6, 1), \rho_1) \quad (26)$$

is determined by the standard differential Chern–Simons refinement together with the fixed convention for $\text{cs}(\rho_1)$. No ITT-specific equivariant-line assumption is needed for this result.

Proof. The differential refinement represents the integral class kc_2 . Evaluation on the closed three-manifold gives the exponentiated Chern–Simons invariant. Substitution of the fixed lens-space value $\text{cs}(\rho_1) = -1/6$ gives the stated phase. $\square$

5 The Induced Cyclic Character

Because $k \in \mathbb{Z}$,

$$Z_k^6 = e^{-2\pi i k} = 1. \quad (27)$$

Therefore, after fixing the generator ζ , the phase defines a genuine cyclic character

$$\boxed{\chi_k^{\text{CS}} : \Gamma \rightarrow U(1), \quad \chi_k^{\text{CS}}(\zeta^n) = Z_k^n.} \quad (28)$$

Thus

$$\boxed{\chi_k^{\text{CS}}(\zeta) = e^{-2\pi i k/6}.} \quad (29)$$

This construction is elementary once Z_k is known. It should not be conflated with a theorem identifying χ_k^{CS} with the fiber character of an independently defined equivariant anomaly line.

Remark 5.1 (Restriction to the selected sector). *The construction above is intentionally attached to the distinguished sector ρ_1 . For general lens-space sectors ρ_n , the Chern–Simons invariant is quadratic in the holonomy label, whereas a group character is linear under powers of a fixed generator. Therefore no general identity of the form $\text{cs}(\rho_n) = \chi(\zeta^n)$ is asserted.*

6 Why the Canonical Equivariant-Line Identification Fails

6.1 Dimension of the standard Chern–Simons line construction

The classical geometric Chern–Simons theory constructs a prequantum line bundle with connection over the moduli space of flat connections on a *two-dimensional* manifold[2]. More generally, the equivariant Chern–Simons bundle construction of Ferreiro Pérez is formulated for compact even-dimensional underlying manifolds and uses equivariant holonomy data[3].

The three-dimensional closed-manifold theory instead assigns a $U(1)$ phase to the full three-dimensional field. Consequently, the expression

$$\mathcal{L}_k \longrightarrow \mathcal{A}(L(6, 1)) \quad (30)$$

as a standard three-dimensional prequantum line bundle is not a legitimate use of the ordinary CS line-bundle construction.

The appropriate three-dimensional datum may be represented by the Chern–Simons phase itself or, in higher-geometric language, by the corresponding invertible/anomaly theory. But this does not automatically produce the desired one-dimensional stabilizer fiber character.

6.2 Equivariant linearization ambiguity

Suppose nevertheless that a one-dimensional complex fiber

$$\mathcal{L}_{k, \rho_1} \quad (31)$$

has been equipped with a genuine Γ -equivariant structure. The generator acts by

$$\widehat{\Phi}_\zeta = \chi_k^{\text{anom}}(\zeta) \text{id}. \quad (32)$$

Let

$$\lambda : \Gamma \rightarrow U(1) \quad (33)$$

be any character. Then

$$\boxed{\widehat{\Phi}'_\gamma = \lambda(\gamma) \widehat{\Phi}_\gamma} \quad (34)$$

is again a genuine equivariant structure. Indeed,

$$\widehat{\Phi}'_{\gamma_1} \widehat{\Phi}'_{\gamma_2} = \lambda(\gamma_1) \lambda(\gamma_2) \widehat{\Phi}_{\gamma_1} \widehat{\Phi}_{\gamma_2} \quad (35)$$

$$= \lambda(\gamma_1 \gamma_2) \widehat{\Phi}_{\gamma_1 \gamma_2} \quad (36)$$

$$= \widehat{\Phi}'_{\gamma_1 \gamma_2}. \quad (37)$$

For $\Gamma = \mathbb{Z}_6$,

$$\text{Hom}(\Gamma, U(1)) \cong \mu_6. \quad (38)$$

Hence the fiber character is not uniquely determined by the underlying line and ordinary differential Chern–Simons class alone.

Proposition 6.1 (Character ambiguity). *The data*

$$(L(6, 1), \rho_1, \widehat{c}_{2,k}) \quad (39)$$

do not uniquely determine a $\mathbb{Z}_6$ -equivariant linearization of an arbitrary one-dimensional fiber. Any proposed character can be modified by a factor in $\text{Hom}(\mathbb{Z}_6, U(1)) \cong \mu_6$ without changing the underlying non-equivariant fiber.

Proof. The construction above explicitly twists an equivariant structure by an arbitrary character λ . The resulting linearization obeys the same group law. Therefore an additional normalization datum is necessary to single out one character. $\square$

6.3 The mapping-torus route does not recover the lens-space phase

One might try to obtain the desired generator action from an ordinary mapping torus. Consider the smooth isotopies

$$\delta_t(z_1, z_2) = \left(e^{2\pi it/6} z_1, e^{2\pi it/6} z_2 \right) \quad (40)$$

and

$$h_t = \begin{pmatrix} e^{2\pi it/6} & 0 \\ 0 & e^{-2\pi it/6} \end{pmatrix}, \quad (41)$$

with $0 \leq t \leq 1$. These give an isotopy of the lifted bundle automorphism from the identity to

$$\Phi_\zeta(x, u) = \left(\delta_\zeta x, \rho_1(\zeta)^{-1} u \right). \quad (42)$$

Thus the associated ordinary mapping-torus construction is isotopic to the product construction on $S^3 \times S^1$ and cannot reproduce the non-zero rational invariant $-1/6$ as an ordinary closed-four-dimensional second Chern number.

This does not contradict the lens-space Chern–Simons invariant. The latter is a three-dimensional secondary invariant of the quotient flat sector, whereas the mapping-torus characteristic number measures a different four-dimensional construction.

Remark 6.2. *The distinction also rules out replacing the rational lens-space invariant by the assertion*

$$\int_{L(6,1) \times S^1} c_2 = -\frac{1}{6}. \quad (43)$$

For an honest closed four-manifold and an honest $SU(2)$ bundle, the ordinary characteristic number is integral.

7 D-01b.3: Precise No-Go Statement

The preceding arguments establish the following restricted conclusion.

Theorem 7.1 (No canonical three-dimensional fiber identification from the present data). *From the specified data*

$$(L(6, 1), \rho_1, \widehat{c}_{2,k}) \quad (44)$$

one can derive the canonical Chern–Simons phase

$$Z_k = e^{-2\pi i k/6} \quad (45)$$

and the induced cyclic character

$$\chi_k^{\text{CS}}(\zeta) = e^{-2\pi i k/6}. \quad (46)$$

However, those data do not by themselves canonically determine an identification

$$\mathcal{L}_{k,\rho_1}^\Gamma \cong \mathbb{C}_{\chi_k^{\text{CS}}} \quad (47)$$

for a hypothetical one-dimensional equivariant fiber over the three-dimensional connection sector. Such an identification requires additional structure or normalization beyond the currently specified differential Chern–Simons data.

Proof. The differential character determines the closed-manifold phase by standard Chern–Simons evaluation. The standard line-bundle construction is a transgression associated with even-dimensional base manifolds, so it does not by itself supply a canonical ordinary line over the three-dimensional connection space. Independently, any one-dimensional equivariant linearization is ambiguous under twisting by an arbitrary $U(1)$ character of Γ . Hence there is no canonical identification with the specific character χ_k^{CS} without additional data. $\square$

8 Revised ITT Selection Principle

The obstruction above does not remove the derived Chern–Simons phase. Instead, it changes the location of the ITT-specific input.

Postulate 8.1 (ITT Fundamental-Sector Phase Triviality). *The fundamental ITT quotient sector is admissible at level k only when*

$$\boxed{Z_k(L(6, 1), \rho_1) = 1.} \quad (48)$$

Using

$$Z_k = e^{-2\pi i k/6}, \quad (49)$$

we obtain

$$e^{-2\pi i k/6} = 1 \iff \frac{k}{6} \in \mathbb{Z}. \quad (50)$$

Thus

$$\boxed{k \in 6\mathbb{Z}.} \quad (51)$$

Corollary 8.2 (Conditional minimal level). *Under ordinary integer Chern–Simons quantization and the ITT fundamental-sector phase-triviality postulate,*

$$\boxed{k_{\min} = 6.} \quad (52)$$

9 Interpretation and Limitations

The mathematical chain established in this paper is

$$\boxed{\text{cs}(\rho_1) = -\frac{1}{6} \implies Z_k = e^{-2\pi i k/6} \implies \chi_k^{\text{CS}}(\zeta) = e^{-2\pi i k/6}.} \quad (53)$$

The next step,

$$\chi_k^{\text{CS}} = 1, \quad (54)$$

is not forced by ordinary Chern–Simons theory. It is the explicit ITT phase-triviality selection postulate.

Accordingly, this paper does not claim a universal derivation of $k = 6$ from $SU(2)$ Chern–Simons theory. It establishes the standard differential-CS phase, isolates the obstruction to the originally proposed canonical equivariant-fiber interpretation, and gives a mathematically unambiguous location for the ITT-specific input.

No argument involving 3×2 , three fermion generations, Standard-Model anomaly counting, or phenomenological parameter fitting is used. The future generation problem remains independent.

10 Conclusion

The principal outcome is a clean separation between what is supplied by standard Chern–Simons geometry and what must be specified by ITT.

First,

$$\text{cs}(\rho_1) = -\frac{1}{6} \quad (55)$$

is a convention-fixed rational invariant of the selected flat lens-space sector. Second, differential Chern–Simons theory derives

$$Z_k = e^{-2\pi i k/6}. \quad (56)$$

Third, this phase defines a cyclic character relative to the chosen generator of $\mathbb{Z}_6$. Fourth, the desired identification of this character with a canonical stabilizer action on a one-dimensional three-dimensional

prequantum fiber is not determined by the present data; the standard transgression dimension and equivariant character ambiguity provide independent obstructions.

The mathematically conservative ITT replacement is the explicit selection principle

$$Z_k(L(6, 1), \rho_1) = 1, \quad (57)$$

which yields

$$\boxed{k_{\min} = 6} \quad (58)$$

as an ITT-specific consequence conditional on that postulate. This is the complete level-selection result established by the present paper.

References

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